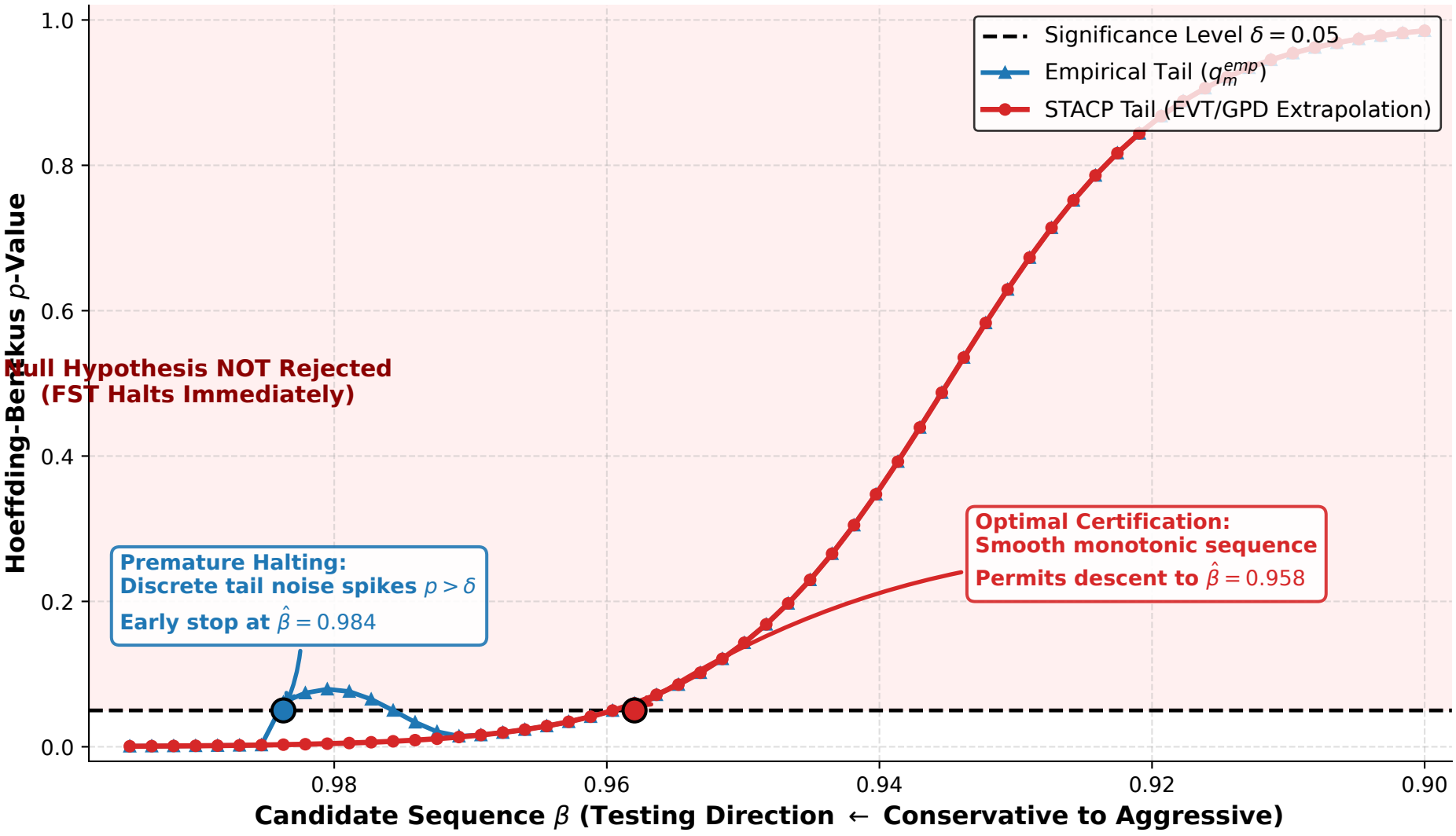
Fixed Sequence Testing (FST) p-Value Trajectory
Significance Level $\delta = 0.05$
Empirical Tail (q_m^{emp})
STACP Tail (EVT/GPD Extrapolation)
Hoeffding–Bentkus p-Value
1.0
0.8
0.6
0.4
0.2
0.0
Null Hypothesis NOT Rejected
(FST Halts Immediately)
Premature Halting:
Discrete tail noise spikes $p > \delta$
Early stop at $\hat{\beta} = 0.984$
Optimal Certification:
Smooth monotonic sequence
Permits descent to $\hat{\beta} = 0.958$
0.98
0.96
0.94
0.92
0.90
Candidate Sequence β (Testing Direction ← Conservative to Aggressive)